

GenAI In Standardization – From Standards to Requirements

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Using generative AI to support standardization work – the case of 3GPP

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Standardization processes build upon consensus between partners, which depends on their ability to identify points of disagreement and resolving them. Large standardization organizations, like the 3GPP or ISO, rely on leaders of work packages who can correctly, and efficiently, identify disagreements, discuss them and reach a consensus. This task, however, is effort-, labor-intensive and costly. In this paper, we address the problem of identifying similarities, dissimilarities and discussion points using large language models. In a design science research study, we work with one of the organizations which leads several workgroups in the 3GPP standard. Our goal is to understand how well the language models can support the standardization process in becoming more cost-efficient, faster and more reliable. Our results show that generic models for text summarization correlate well with domain expert’s and delegate’s assessments (Pearson correlation between 0.66 and 0.98), but that there is a need for domain-specific models to provide better discussion materials for the standardization groups.

CCS Concepts: • **Social and professional topics** → **Testing, certification and licensing**.

Additional Key Words and Phrases: machine learning, standardization, requirements, automation

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1 INTRODUCTION

Standardization in software and systems engineering is typically spearheaded by industry-led organizations such as the 3rd Generation Partnership Project (3GPP), the International Standards Organization (ISO) or Automotive AUTOSAR consortium. The development of new standards, or the modification of existing ones, relies heavily on reaching a consensus among member entities. Usually, the member organizations provide input before meetings, which needs to be read, organized, summarized, and then presented at the meetings [Baron et al. 2019; Baron and Gupta 2018].

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In this context, sizable standardization companies face the time-consuming and effort-intensive task of analyzing a significant volume of documents that must be read, distilled, and showcased during discussions. Achieving consensus, while also delivering tangible benefits to the contributing bodies, necessitates an effective analysis of these submissions. A more streamlined process not only results in superior standards but also ensures a more effective use of the experts’ time who are integral to the standardization journey. In the end, such a process has a significant impact on the society as a whole [Bruer and Brake 2021].

Currently, the analysis is mostly a manual task, with support only in terms of contribution change-tracking, but this can be improved with the use of Large Language Models (LLMs). We observe the rise of new LLMs that can help in the analysis of these large documents. Therefore, in this paper, we address the following research question:

To which degree can large language models provide relevant summaries of standardization documents?

In particular, we set off to analyze the BART model and the Pegasus XLM models for the following tasks:

- (1) summarization of large documents – with the purpose of getting a quick orientation of the main contributions,
- (2) similarity analysis – with the purpose of identifying the features which the members agree on (and where more discussions are needed), and
- (3) overlap analysis – with the purpose of guiding the discussions in smaller groups of companies.

We use the design science research approach [Wieringa 2014], where the artifact is the machine learning-based system for supporting the standardization processes. We evaluate it with our industrial partner Ericsson AB in the context of their past 3GPP RAN standardization activities, following our previous studies [Ochodek et al. 2022].

The rest of the paper is structured as follows. Section 2 outlines the most relevant related research studies in the area of text summarization and standardization efforts. Section 3 presents the details of our research design. Section 4 presents the artefact – standards document summarization and analysis system – and Section 5 presents the results of its evaluation. Section 6 discusses the results and the validity of our study. Finally, Section 7 presents the conclusions.

2 RELATED WORK

A lot of current research effort related to 3GPP standardization goes into the development of standards for UAVs (drones) [Abdalla and Marojevic 2021], where telecommunication plays a crucial role; in particular low latency communication. Standardization is important for all kinds of devices, not only drones [Kar and Sanyal 2020]. The number of such standards indicates how much-automated support

Outline

- 3GPP standardization – consensus development
- GenAI in standization – a use case and results
- GenAI for requirements generation
- Based on M. Staron, J. Ström, A. Karlsson and W. Meding, *Using generative AI to support standardization work – the case of 3GPP*, 50th international conference on Software Engineering and Advanced Applications (SEAA), 2024

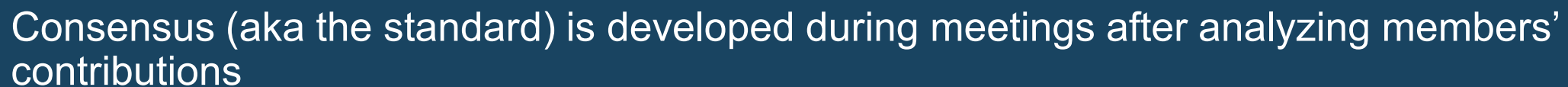


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Issue 4: the final text may take a long time to author and agree upon



Example of a discussion item

Based on the members' proposal the leader creates a proposal for the standard

Some companies agree, while some require modifications

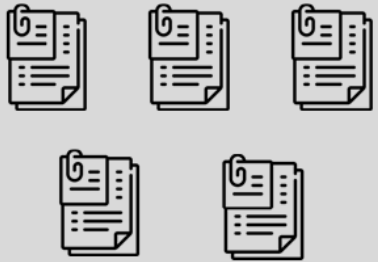
Modifications may be “small” in terms of a text, but “huge” in terms of the impact

FL1 High Priority Proposal 2.2-1: Replace the RAN1#104bis-e working assumption with the following agreement:

- After initial access (i.e., after RRC Setup, RRC Resume, or RRC Reestablishment), for both BWP#0 configuration options 1 and 2 (as in 38.331, Appendix B2), a RedCap UE is not expected to operate with an initial DL BWP wider than the maximum RedCap UE bandwidth.

Company	Y/N	Comments
██████████	Y	
██████████	Y	
██████████	Y	
██████████	Y	
██████████	Y	
██████████	Y, but	We are fine to go this direction, but design should ensure that <u>gNB</u> provides an non-cell-defining SSB (i.e. SSB without PBCH) in initial DL BWP used after initial access, and needed at least for serving cell RRM.
██████████	Y	After initial access, it is natural that <u>gNB</u> should configure the initial DL BWP no wider than the <u>RedCap UE</u> bandwidth. After the effective time of RRC reconfiguration, it is natural that <u>gNB</u> should configure the BWP (including the initial DL BWP) no wider than the <u>RedCap UE</u> bandwidth. There is no spec impact.

Using GenAI to address these problems



Member
contributions

LLMs are used to summarize the text in each contribution

- Section by section

Cosine distances are used to find similarities

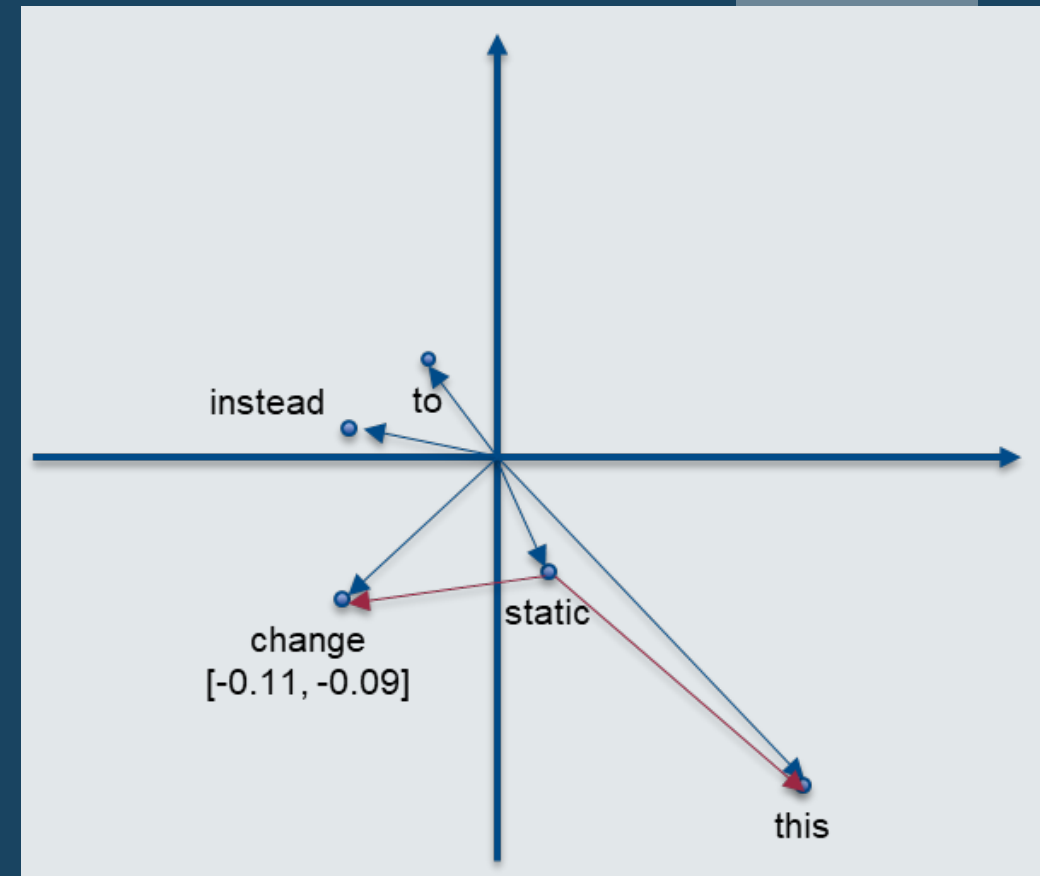
- Section by section, heading by heading

This gives us the agreements

Underlying principle similarity == distance

Tokens:	change	this	to	static	instead
Embeddings vectors:	$\begin{bmatrix} -0.11 \\ \dots \\ -0.09 \end{bmatrix}$	$\begin{bmatrix} 0.1 \\ \dots \\ -0.25 \end{bmatrix}$	$\begin{bmatrix} -0.01 \\ \dots \\ 0.05 \end{bmatrix}$	$\begin{bmatrix} 0.06 \\ \dots \\ -0.04 \end{bmatrix}$	$\begin{bmatrix} -0.02 \\ \dots \\ 0.02 \end{bmatrix}$

We can plot these vectors on an n-dimensional space and use vector arithmetic.



$$d\left(\begin{bmatrix} -0.11 \\ -0.09 \end{bmatrix}, \begin{bmatrix} 0.06 \\ -0.04 \end{bmatrix}\right) = 0.1772$$

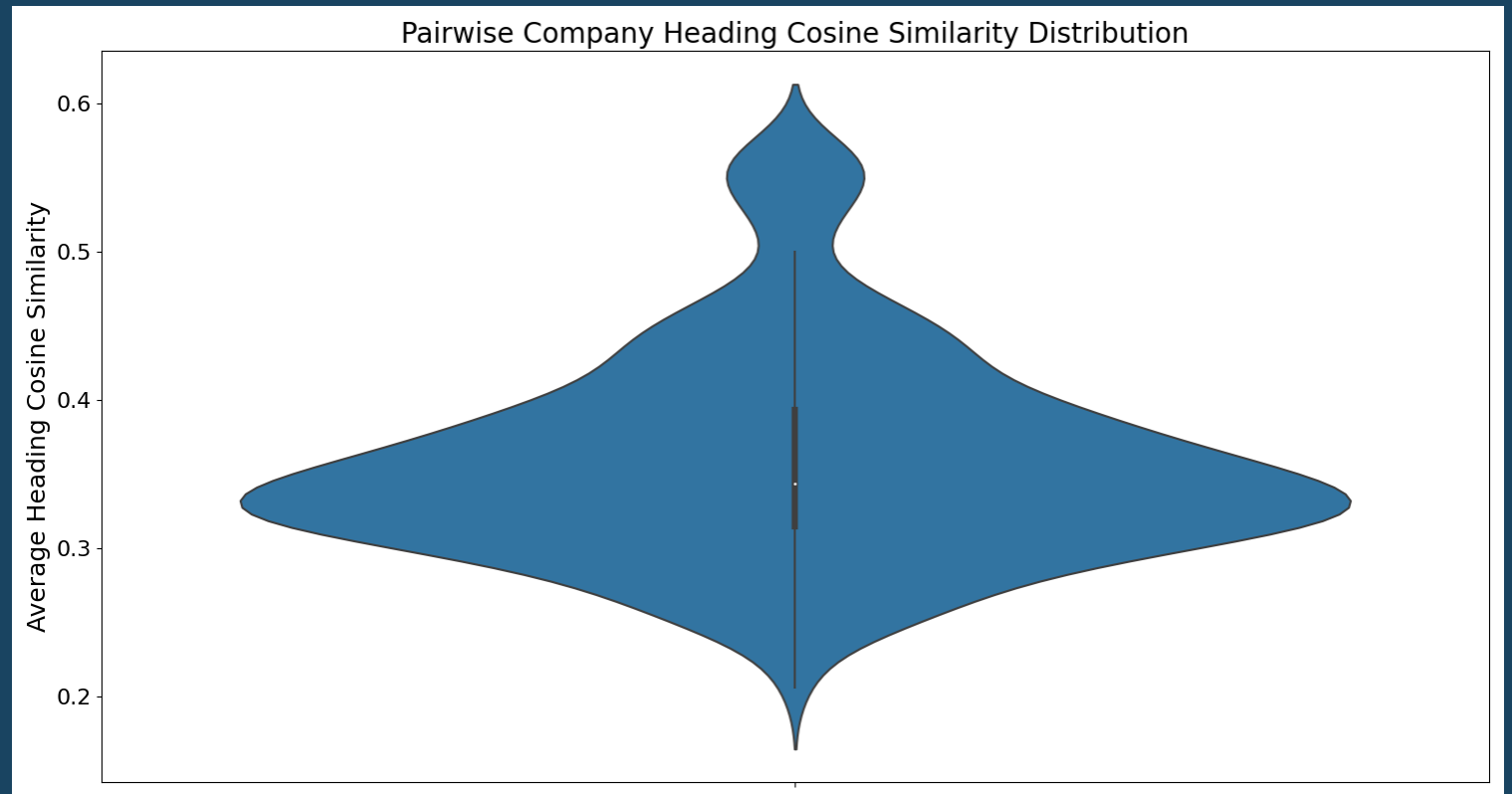
Please note that I'm using a simplification variant here, for pedagogical reasons – instead of n-dimensions, I only use two. In practice I use t-SNE transformation to transform n-dimensions into two/three for plotting

Understanding the general status of the standard

Pairwise similarity shows whether the companies take up the same issues in their contributions

Low similarity == a lot to talk about

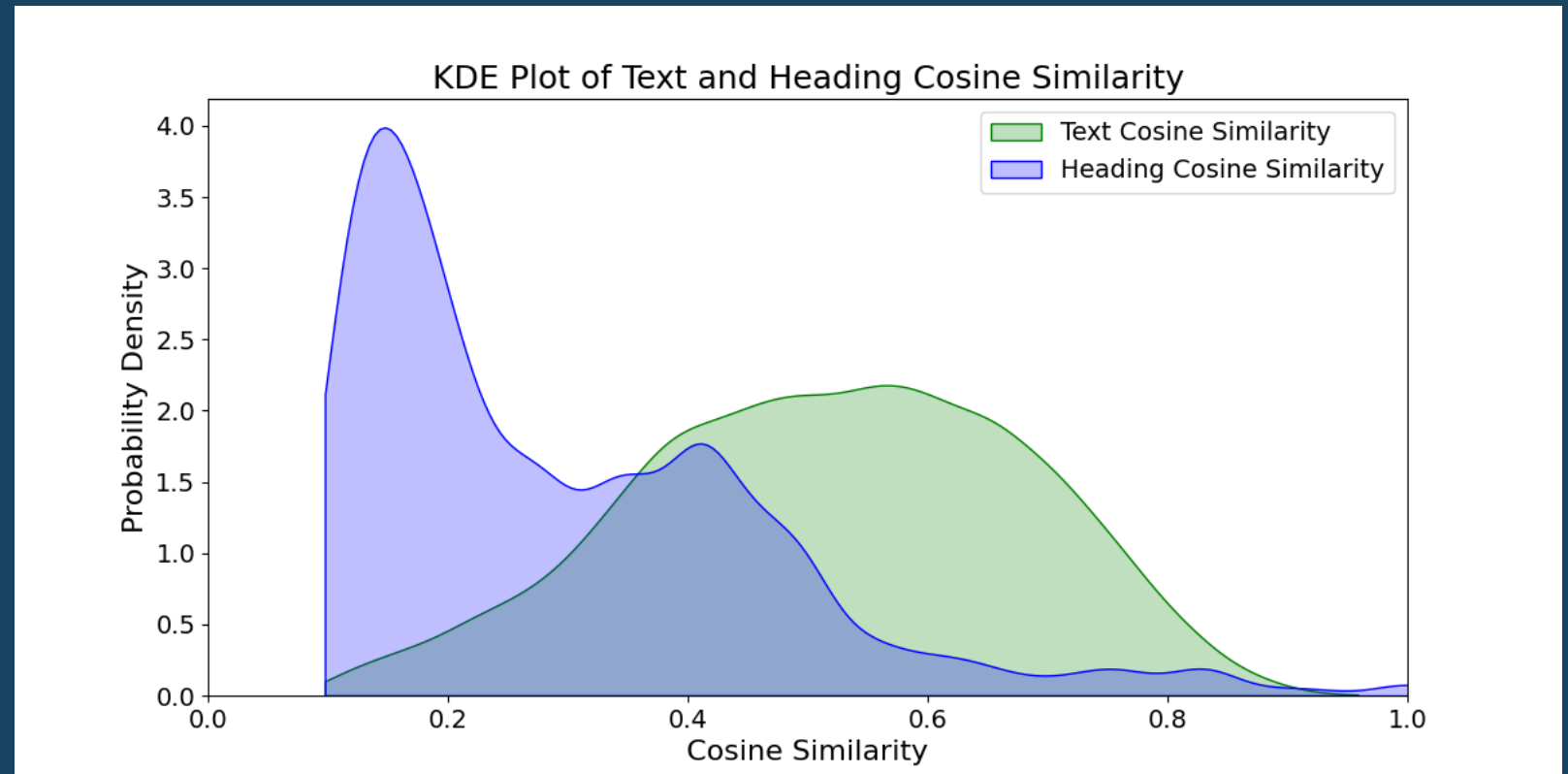
High similarity == ready for consensus/standardization



Understanding the general status of the standard

Headings → members' own view on the topics

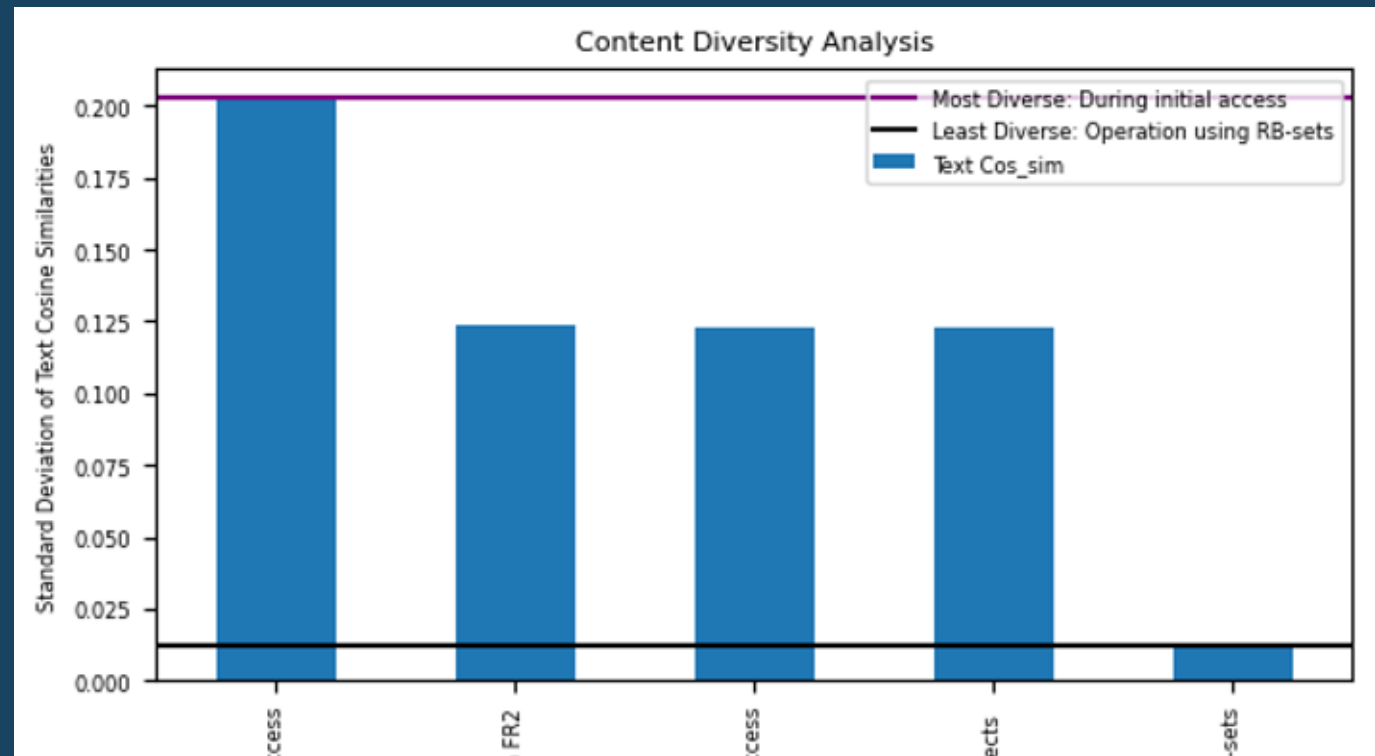
Content → Detailed differences in the topics



Finding contributions that are similar or different

Now that we have the similarity analysis, we can move to the next step

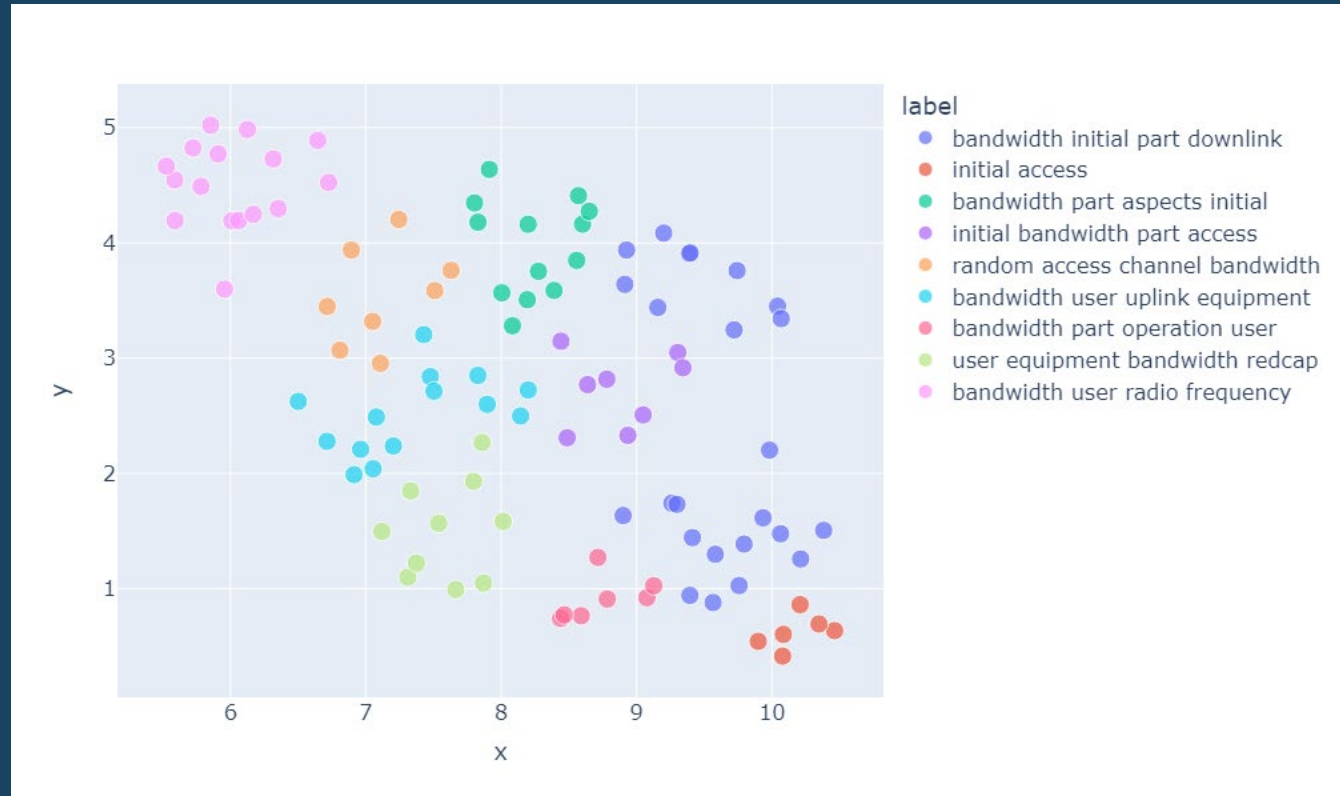
What are the differences and similarities?



Agreement or disagreement topics

Using the t-SNE diagrams we get an understanding of groups of topics (sections) that the members agree or disagree upon

By using summarization we can set a label on these items == helps in prioritization

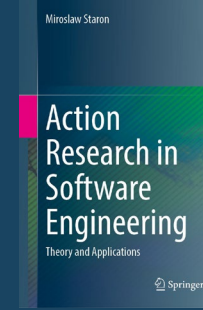
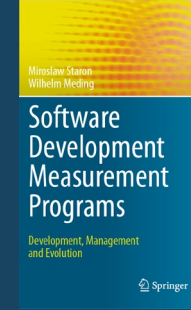
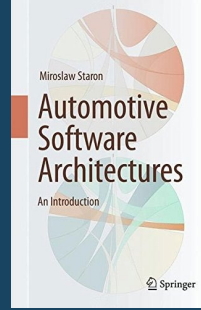


Finally – an agenda with discussion points

In the end we generate a proposed agenda for the meeting.

The moderator/leader can use it to create the final agenda and lead the meeting more efficiently

1 Introduction This feature lead (FL) summary (FLS) concerns the Rel-17 work item (WI) for support of reduced capability (RedCap) NR devices [1]. Earlier RAN1 agreements for this WI are summarized in [2]. This document summarizes contributions [3] – [31] submitted to agenda item 8.6.1.1 and relevant parts of contributions [32] – [34] submitted to agenda item 8.6.3 and captures this email discussion on reduced maximum UE bandwidth: [105-e-NR-R17-RedCap-01] Email discussion regarding aspects related to reduced maximum UE bandwidth – Johan (Ericsson) 1st check point: 521 2nd check point: 523 Final check: 521 The final FLS from the previous RAN1 meeting and the draft LS that was discussed then can be found in [35] and [36]. The issues in this document are tagged and color coded like this: 1. High Priority 2. Medium Priority The following proposals are candidates for treatment in online (GTW) session: High Priority Proposal 2.1.1: Confirm the following RAN1#104bis-e working assumption: • During initial access, the bandwidth of the initial DL BWP for RedCap UEs is not expected to exceed the maximum RedCap UE bandwidth. • The bandwidth and location of the initial DL BWP for RedCap UEs can be the same as the bandwidth and location of the MIB-configured initial DL BWP for non-RedCap UEs. • This does not preclude a SIB-configured initial DL BWP for non-RedCap UEs only with a wider bandwidth than the maximum RedCap UE bandwidth. • This does not preclude separate or additional bandwidth and location for initial DL BWP for RedCap UEs (FFS). High Priority Proposal 2.1.2a: • An initial DL BWP for RedCap UEs for use during initial access can be configured separately from the initial DL BWP for non-RedCap UEs. • The same MIB-configured CORESET#0 is assumed and additional CORESETs are FFS. High Priority Proposal 2.2.1a: Replace the RAN1#104bis-e working assumption with the following revised working assumption:	• Working assumption: After initial access (i.e., after RRC Setup, RRC Resume, or RRC Reestablishment), for both BWP#0 configuration options 1 and 2 (as in 38.331, Appendix B2), a RedCap UE is not expected to operate with an initial DL BWP wider than the maximum RedCap UE bandwidth. High Priority Proposal 2.2.2: • If an initial DL BWP for RedCap UEs for use during initial access is configured separately from the initial DL BWP for non-RedCap UEs, this separately configured initial DL BWP for RedCap UEs can also be used after initial access (i.e., after RRC Setup, RRC Resume, or RRC Reestablishment). High Priority Proposal 3.1.1a: • Both during and after initial access, the scenario where the initial UL BWP for non-RedCap UEs is configured to be wider than the maximum RedCap UE bandwidth is allowed. High Priority Proposal 3.1.2a: • Working assumption: Both during and after initial access, for the scenario where the initial UL BWP for non-RedCap UEs is configured to be wider than the RedCap UE bandwidth, a separate initial UL BWP no wider than the RedCap UE maximum bandwidth is configured/defined for RedCap UEs. • Strive for a mean to avoid or minimize the PUSCH resource fragmentation due to PUCCH transmission for the above case. • It is FFS whether/when the centre frequencies for initial DL and UL BWPs can be different in TDD. High Priority Proposal 4.1: Confirm the main bullet of the RAN1#104bis-e working assumption, i.e.: • A RedCap UE cannot be configured with a non-initial (DL or UL) BWP (i.e., a BWP with a non-zero index) wider than the maximum bandwidth of the RedCap UE.
2 Initial DL BWP	2.1 Initial DL BWP during initial access
RAN1#104bis-e agreed the following working assumption related to initial DL BWP during initial access: Working assumption • During initial access, the bandwidth of the initial DL BWP for RedCap UEs is not expected to exceed the maximum RedCap UE bandwidth. • The bandwidth and location of the initial DL BWP for RedCap UEs can be the same as the bandwidth and location of the MIB-configured initial DL BWP for non-RedCap UEs. • This does not preclude a SIB-configured initial DL BWP for non-RedCap UEs only with a wider bandwidth than the maximum RedCap UE bandwidth. • This does not preclude separate or additional bandwidth and location for initial DL BWP for RedCap UEs (FFS). Regarding the initial DL BWP, contributions unanimously agree to confirm the working assumption indicating that, during initial access, the bandwidth of the initial DL BWP for RedCap UEs is not expected to exceed the maximum RedCap UE bandwidth [3, 4, 6, 7, 9, 10, 13, 14, 17, 18, 19, 21, 22, 26]. However, one contribution [31] discusses that a RedCap UE can be configured with a BWP larger than its maximum supported bandwidth. FL1 High Priority Proposal 2.1.1: Confirm the following RAN1#104bis-e working assumption: • During initial access, the bandwidth of the initial DL BWP for RedCap UEs is not expected to exceed the maximum RedCap UE bandwidth.	



Large Language Models for Requirements analysis

The default QoS Rule shall have the highest possible precedence value, i.e., it has the lowest precedence.

Requirement written by human, telecom expert, who wrote this requirement based on the 3GPP standard



The default QoS Rule shall have the highest possible precedence value, and shall always be applied first.

What is the type of the default QoS Rule?

The default QoS Rule is of type "Default".

Requirement written by openAI based on 40% of the prompt from the expert



2 experts judged this requirement to be somewhat similar to the original one.

The default QoS Rule shall have the highest possible priority, so that it has the higher priority than the QoS Rules with user-defined priority.

Requirement written by openAI based on 50% of the prompt from the expert

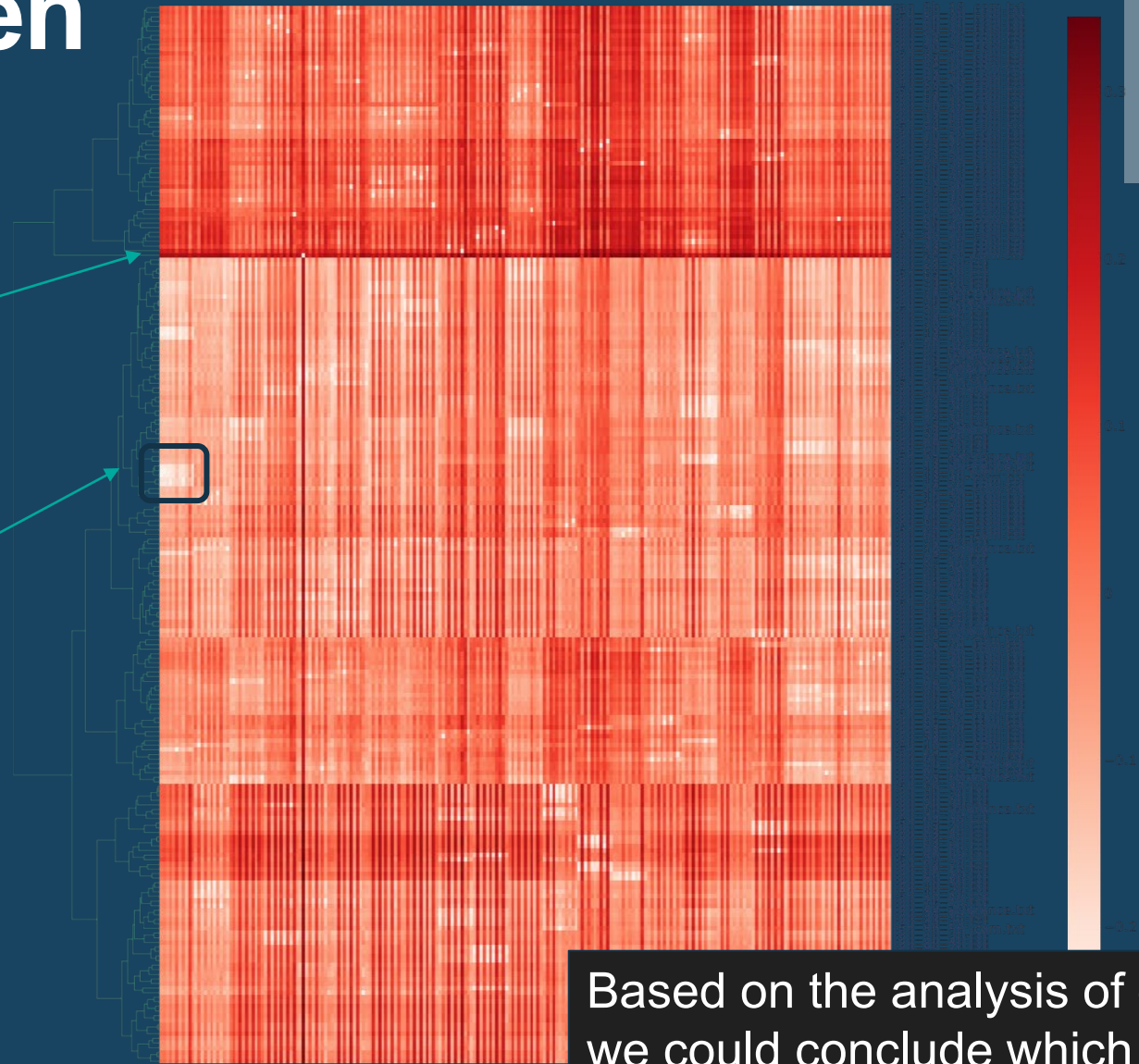


2 experts judged this requirement to be almost identical in meaning with the original requirement.

Distance between requirements

This requirement is very different from all the other ones.

This is a cluster of requirements that are similar == redundant

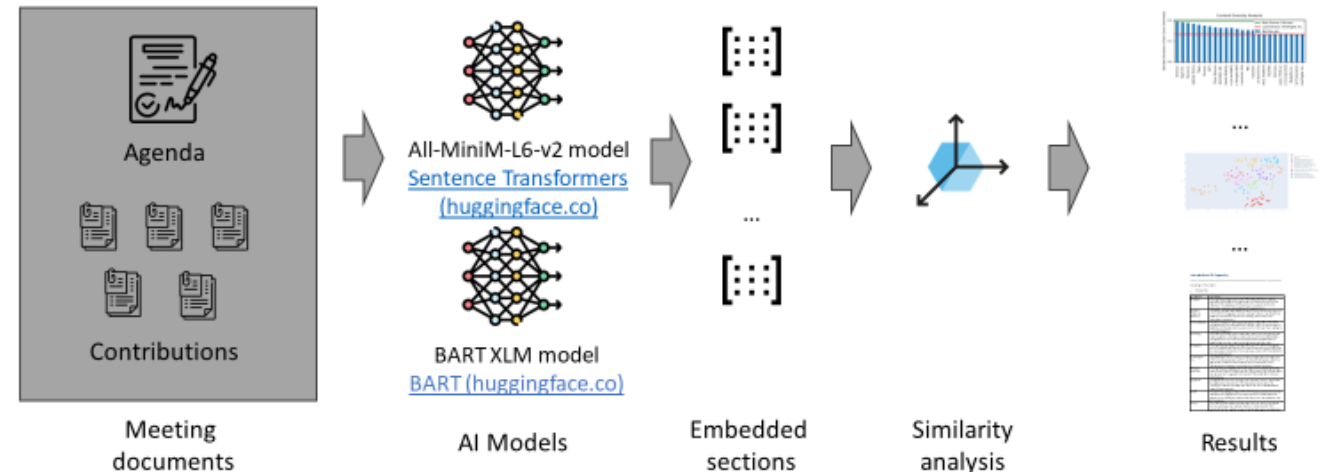


Based on the analysis of distances, we could conclude which of the requirements are redundant and which are "new" compared to the rest

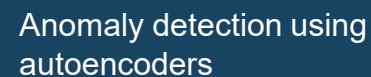
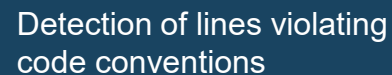
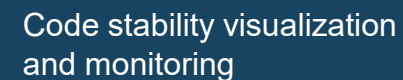
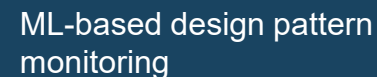
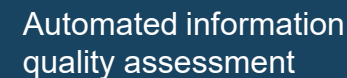
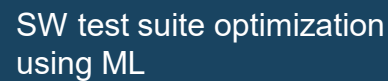
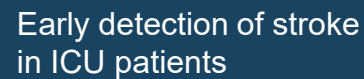
Conclusions

- Using GenAI speeds up the standardization process
 - Days to minutes
- Standardization is only a beginning
 - Writing requirements is the next step
- ISO, IEEE, AUTOSAR work in the same way

ML pipeline in these use cases



Catalogue of best metrics



ML-based extracting SW requirements from 5G standards

